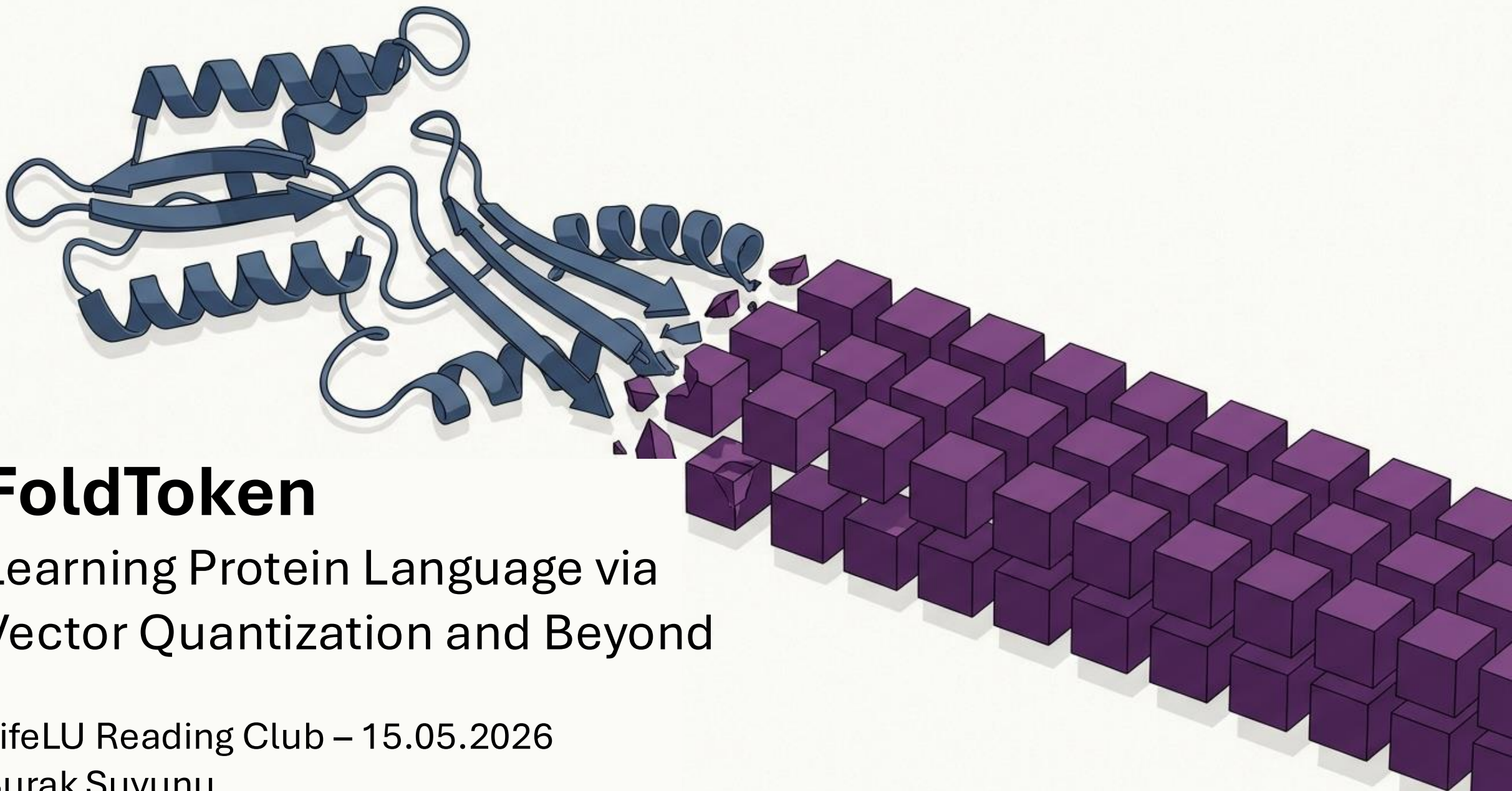




FoldToken

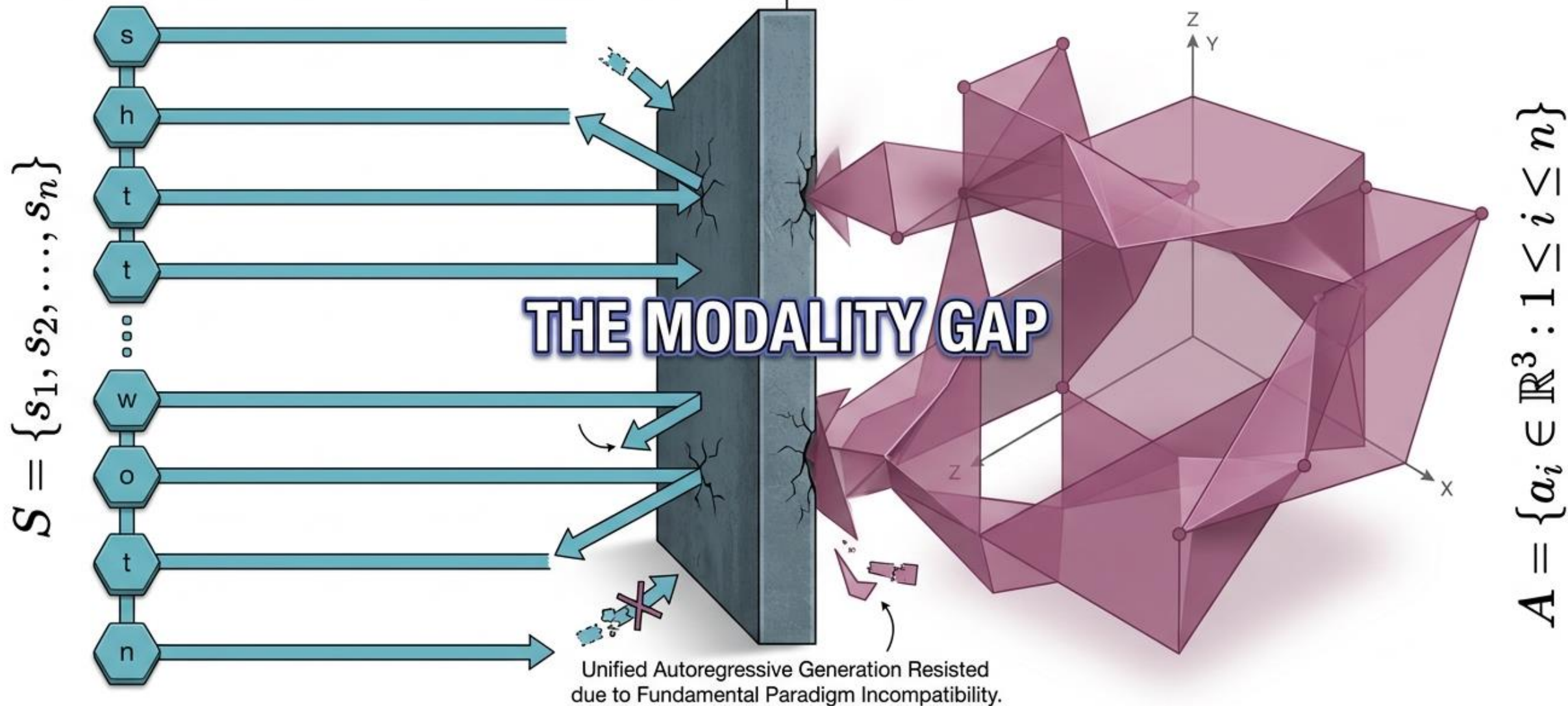
Learning Protein Language via
Vector Quantization and Beyond

LifeLU Reading Club – 15.05.2026
Burak Suyunu

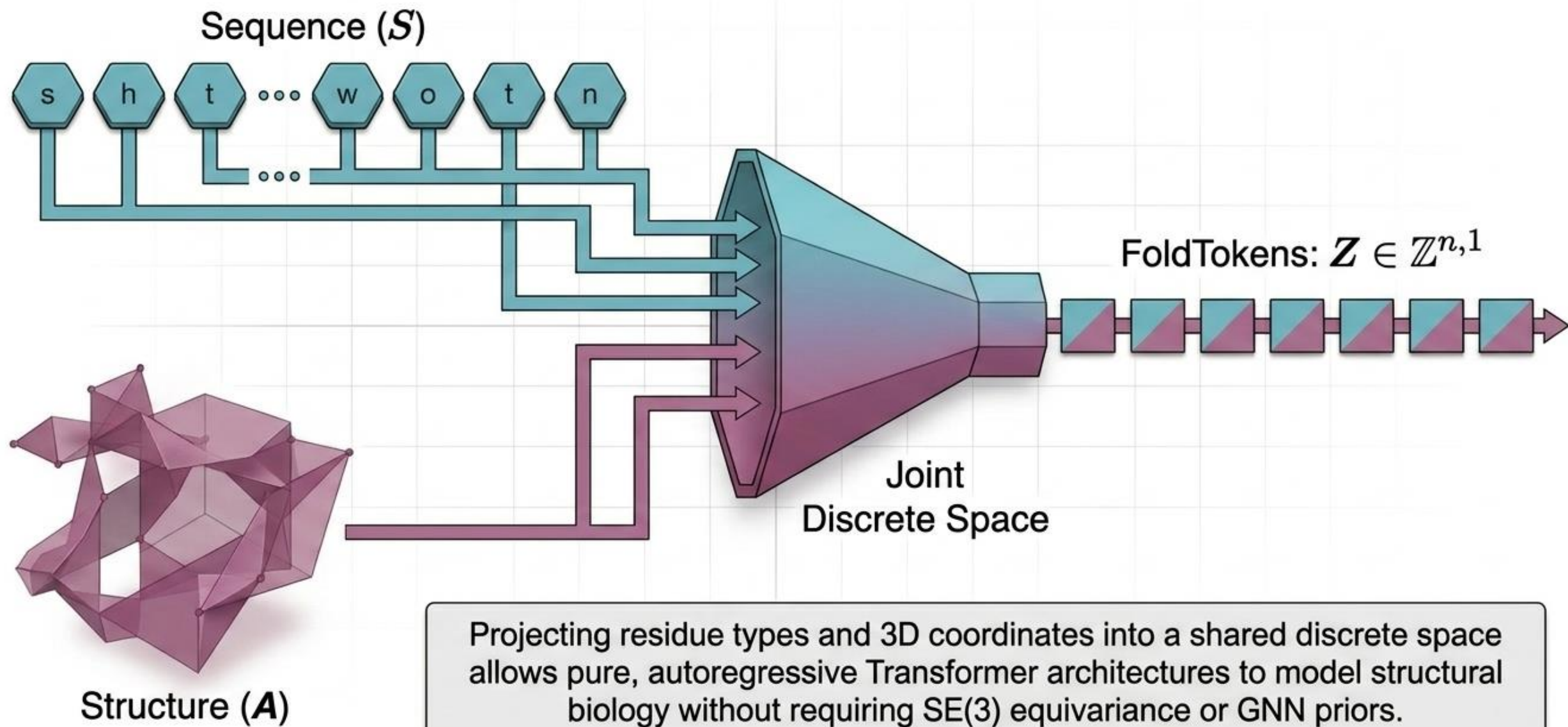
The Modality Gap: Discrete Text vs. Continuous Geometry

1D Discrete Domain / NLP Paradigm / Transformers

3D Continuous Domain / Graph Paradigm / GNNs



The FoldToken Paradigm: A Unified Multimodal Language



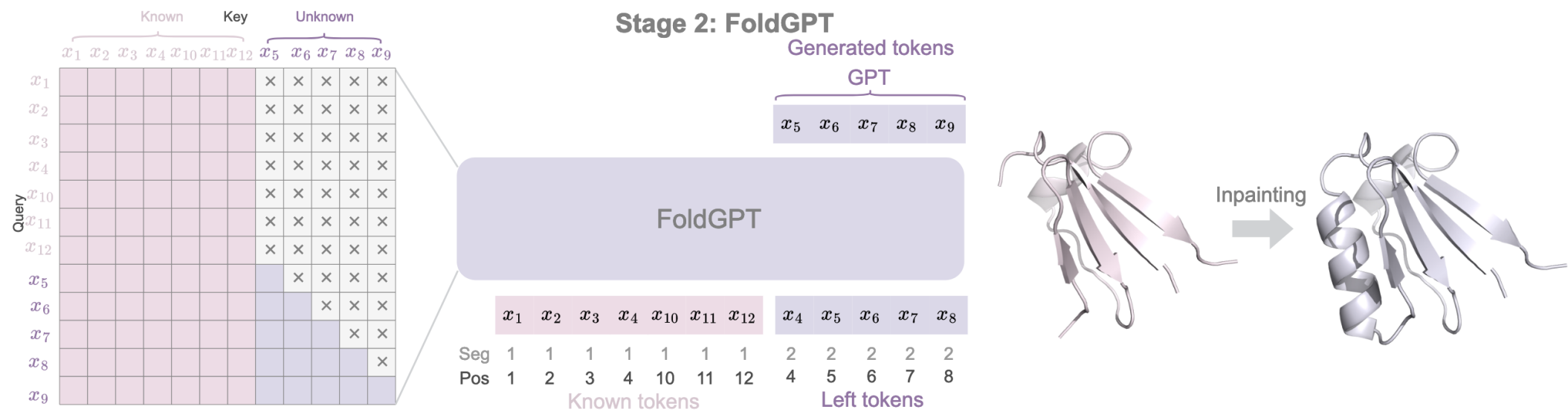
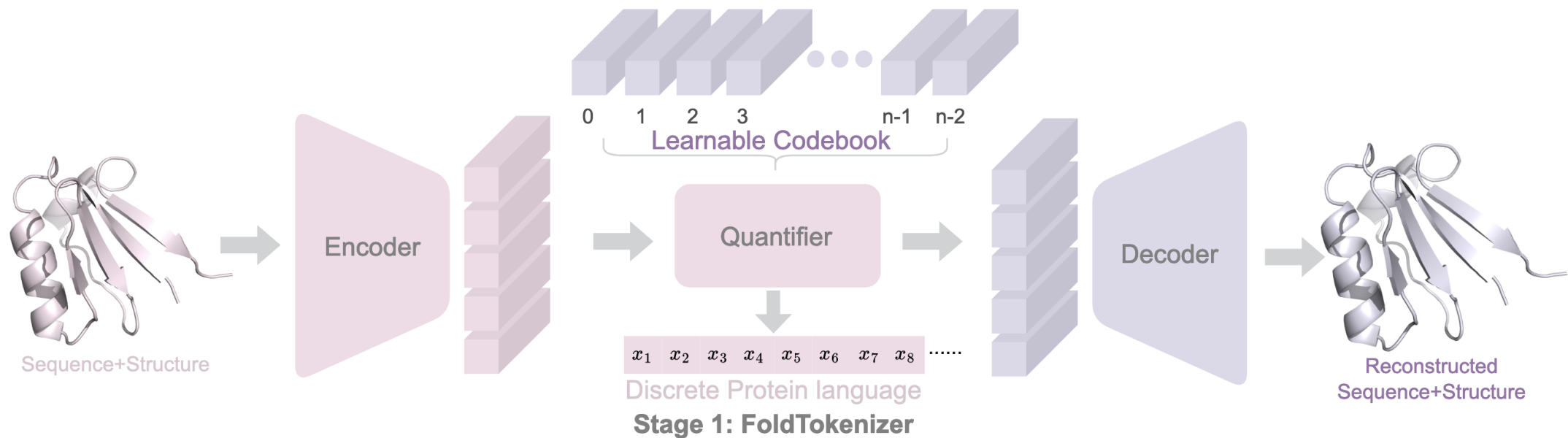
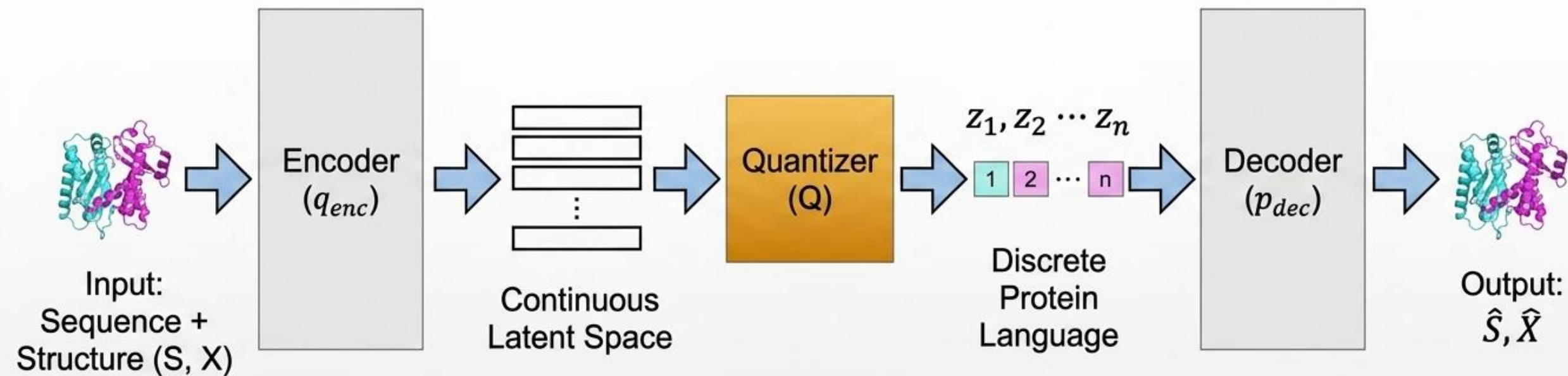


Figure 1: The overall framework of FoldToken.

FoldTokenizer: The Autoencoder Architecture

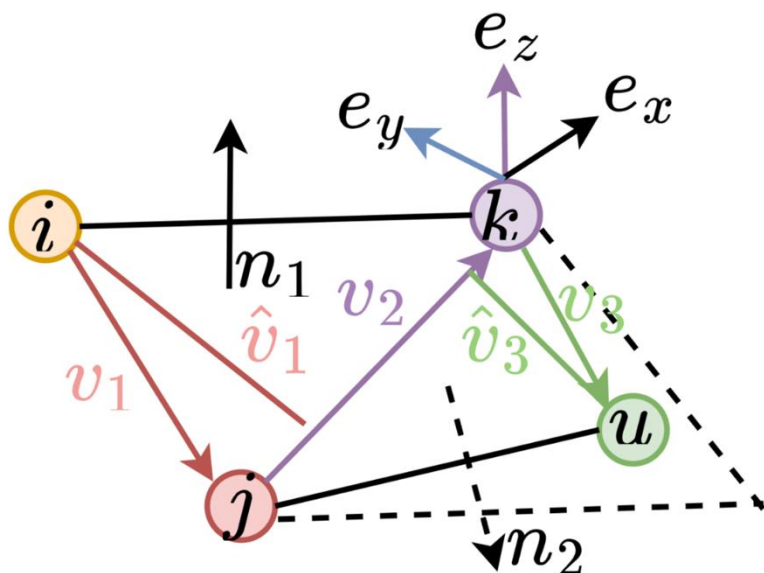


$$\mathbf{h} \sim q_{enc}(\mathbf{h} | S, A)$$

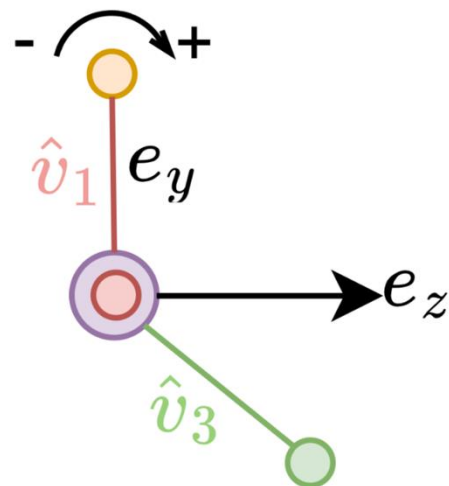
$$\mathbf{z} = Q(\mathbf{h})$$

$$\hat{S}, \hat{A} = p_{dec}(S, A | Q^{-1}(\mathbf{z}))$$

Geometric Representation



(a) Place u based on the plane of i - j - k



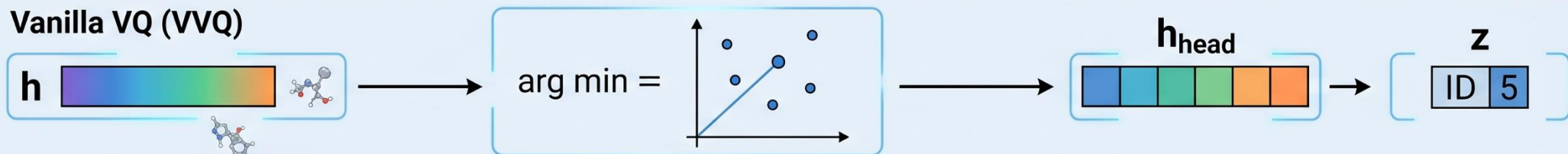
(b) Torsion angle

$$\left\{ \begin{array}{l} \mathbf{v}_1 = \mathbf{x}_j - \mathbf{x}_i \\ \mathbf{v}_2 = \mathbf{x}_k - \mathbf{x}_j \\ \mathbf{v}_3 = \mathbf{x}_u - \mathbf{x}_k \\ \hat{\mathbf{v}}_1 = -\mathbf{v}_1 - ((-\mathbf{v}_1) \cdot \mathbf{v}_2) \mathbf{v}_2 \\ \hat{\mathbf{v}}_3 = \mathbf{v}_3 - (\mathbf{v}_3 \cdot \mathbf{v}_2) \mathbf{v}_2 \\ r_u = \|\mathbf{v}_3\|_2 \\ \alpha_u = \angle(-\hat{\mathbf{v}}_2, \hat{\mathbf{v}}_3) \\ \beta_u = \angle(\hat{\mathbf{v}}_1, \hat{\mathbf{v}}_3) = \angle(\mathbf{n}_1, \mathbf{n}_2) \end{array} \right.$$

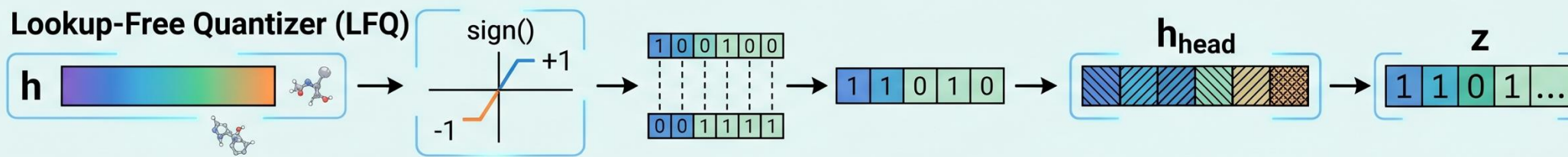
Following the backbone direction, given previous three atoms, we can construct a local frame $T_u = \{\mathbf{x}_i, \mathbf{x}_j, \mathbf{x}_k\}$. The next atom $\mathbf{x}_u$ can be represented by the distance r_u and two torsion angles α_u and β_u , denoted as $\mathbf{a}_u = (r_u, \alpha_u, \beta_u)$.

FoldTokenizer VQ Options

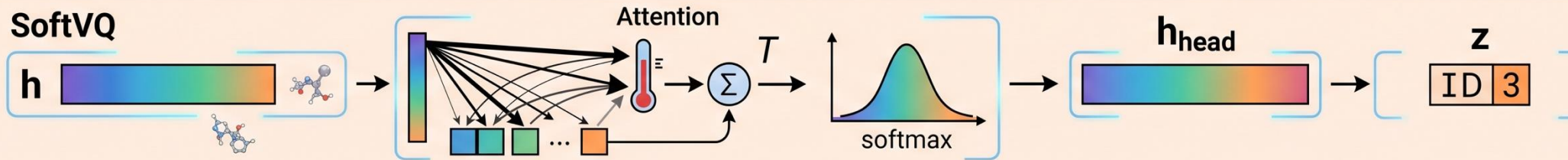
Vanilla VQ (VVQ)



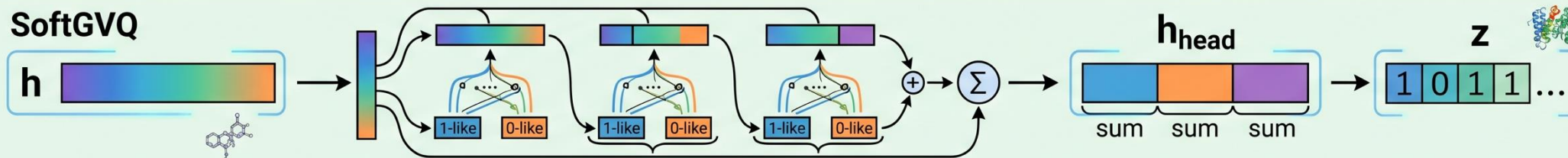
Lookup-Free Quantizer (LFQ)



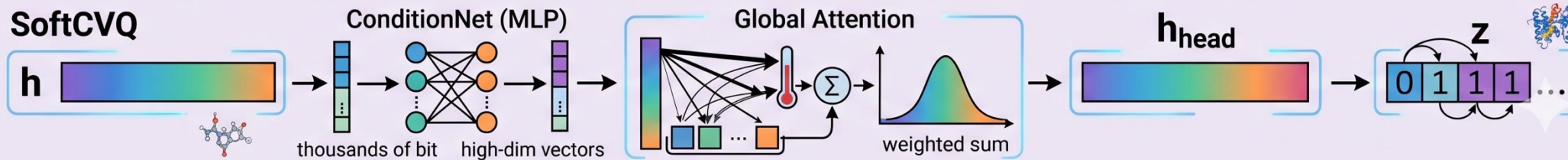
SoftVQ



SoftGVQ

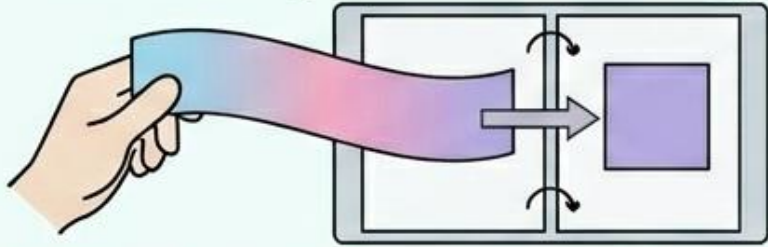


SoftCVQ



VQ Evolution I: Vanilla Quantizer (VVQ)

The Layman Analogy



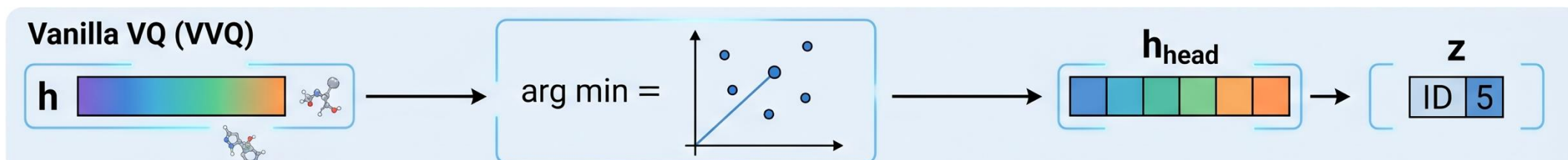
Finding the closest matching paint swatch in a finite book.

Technical Definition

Nearest-Neighbor Mapping.

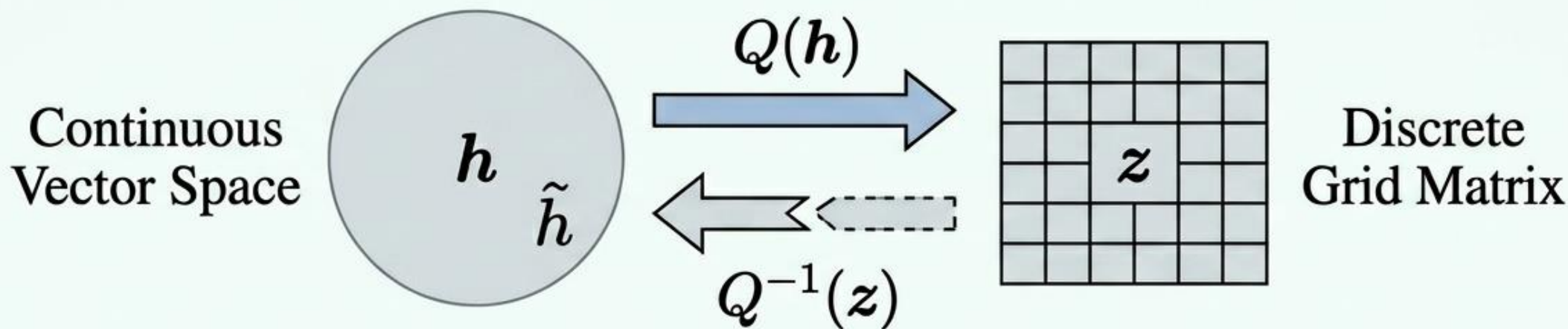
Projects continuous latent representations directly to the single nearest static vector in a pre-defined, fixed codebook.

VQ Evolution I: Vanilla Quantizer (VVQ)

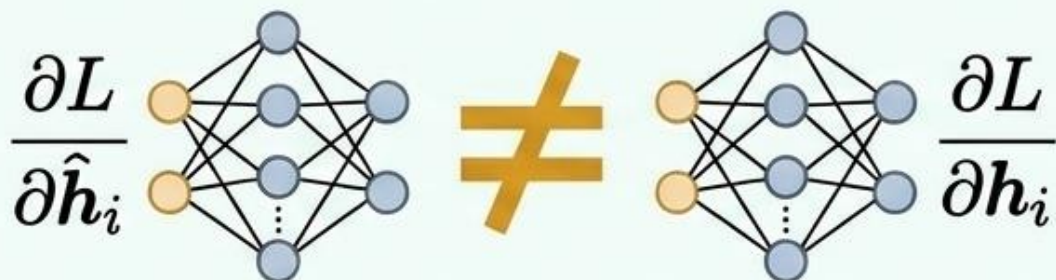


$$\begin{cases} z_i &= Q(\mathbf{h}_i) = \arg \min_j \|\mathbf{h}_i - \mathbf{v}_j\|_2 \\ \hat{\mathbf{h}}_i &= Q^{-1}(z_i) = \mathbf{v}_{z_i} \end{cases}$$

The Quantization Bottleneck



Gradient Mismatch



VQ-IDs are **non-differentiable**. Copying gradients bypasses actual embedding differences, **destroying precise 3D** spatial orientations.

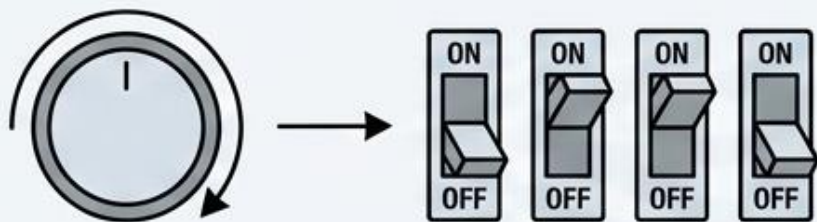
Semantic Irrelevance



Discrete index IDs possess **no ordinal correlation** to their spatial geometric **distance**. The generative model is forced to rote-memorize a vast, unstructured class space

VQ Evolution II: Lookup-Free Quantizer (LFQ)

The Layman Analogy



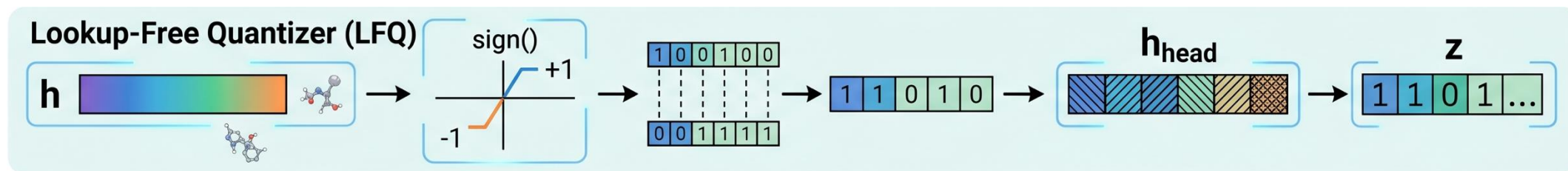
Turning a continuous dial into a series of rigid, binary switches.

Technical Definition

Binary Sign Space Projection.

Eliminates the fixed lookup table entirely. Projects continuous embeddings directly into a binary sign space, creating discrete IDs derived purely from the data's own binary signature to fix semantic irrelevance.

VQ Evolution II: Lookup-Free Quantizer (LFQ)



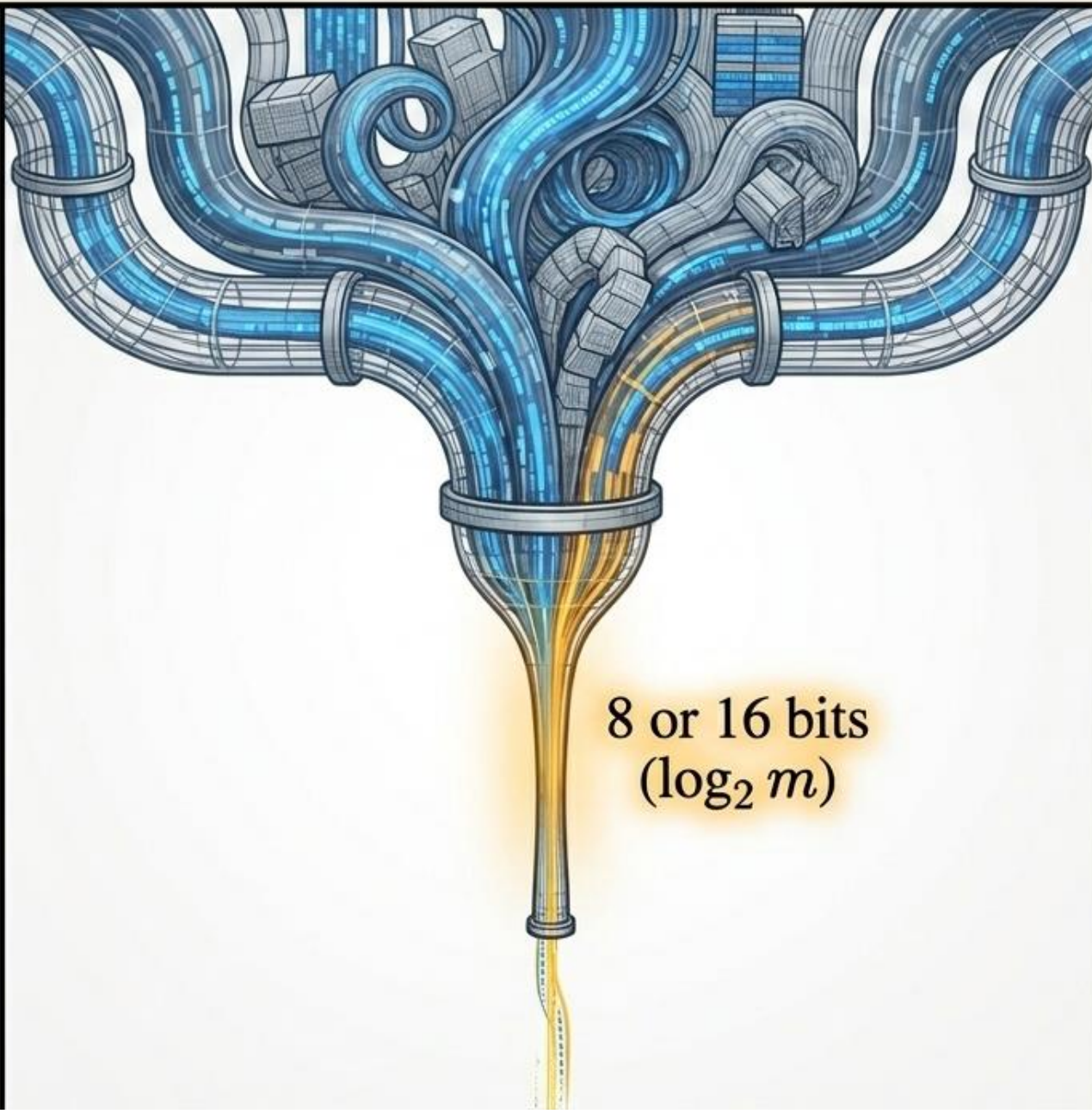
$$\begin{cases} \hat{\mathbf{h}}_i = \text{sign}(\mathbf{h}_i) = \sum_{j=1}^d [u(h_{ij})\mathbf{e}_j + (1 - u(h_{ij}))(-\mathbf{e}_j)] \\ \text{Bit}^{-1}(\mathbf{z}, d) := \sum_{j=1}^d 2^{j-1} z_j \\ \text{Bit}(\mathbf{z}, d) := [\lfloor z/2^{d-1} \rfloor, \lfloor z/2^{d-2} \rfloor, \dots, \lfloor z/2^0 \rfloor]^T \\ z_i = \text{Bit}^{-1}(u(\hat{\mathbf{h}}_{i,j})) \end{cases}$$

(11)

The Information Bottleneck of LFAQ

The Fix: Semantic Relevance

LFAQ successfully creates semantic meaning. Because tokens are binary codes, partial binary matches still carry inherent structural similarity, making autoregressive prediction exponentially easier.



The Failure: Catastrophic Fidelity Loss

To maintain parity with standard vocabulary sizes (m), the hidden representation space is mathematically constrained to $\log_2 m$.

A 16-bit space is drastically too low to compress the vast complexity of 3D protein folding angles, destroying structural reconstruction.

VQ Evolution III: Soft Vector Quantizer (SoftVQ)

The Layman Analogy



Blending multiple swatches based on similarity, slowly narrowing focus to the absolute best match over time.

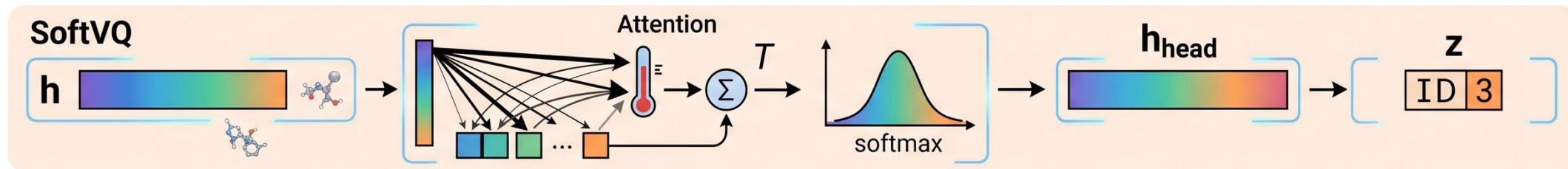
Technical Definition

Differentiable Soft Attention.

Queries the entire codebook space simultaneously with a decaying temperature parameter.

This global querying allows perfectly smooth gradient flow during training, solving gradient mismatch.

VQ Evolution III: Soft Vector Quantizer (SoftVQ)



$$\begin{cases} a_{ij} &= \frac{\exp(\mathbf{h}_i^T \mathbf{v}_j / T)}{\sum_{j=1}^m \exp(\mathbf{h}_i^T \mathbf{v}_j / T)} \\ \hat{\mathbf{h}}_i &= \sum_{j=1}^m a_{ij} \mathbf{v}_j \\ z_i &= \arg \max_j a_{ij} \end{cases}$$

VQ Evolution IV: Soft Group Vector Quantizer (SoftGVQ)

The Layman Analogy



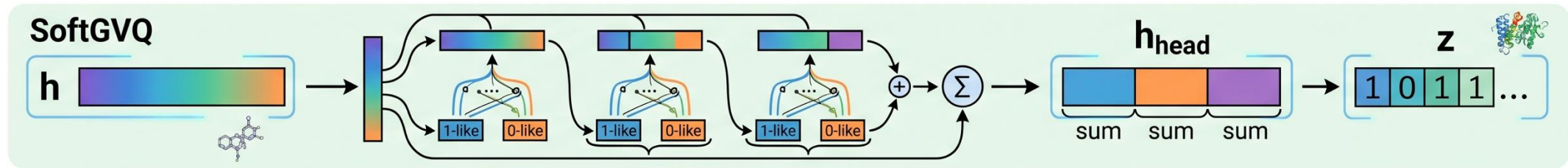
Grouping continuous dials and blending within isolated groups to capture high-resolution details in a binary format.

Technical Definition

Grouped Soft Attention + Learnable Basis.

Combines soft attention with learnable binary basis vectors, separated into distinct groups. This completely bypasses the LFQ dimension bottleneck while maintaining sequence-friendly binary IDs.

VQ Evolution IV: Soft Group Vector Quantizer (SoftGVQ)



Mathematical Derivation

$$w_{ij} = \frac{\exp(h_i^T v_j / T)}{\exp(h_i^T v_j / T) + \exp(h_i^T \dot{v}_j / T)}$$

$$\hat{h}_i = \sum [w_{ij} v_j + (1 - w_{ij}) \dot{v}_j]$$

VQ Evolution V: Soft Conditional Vector Quantizer (SoftCVQ)

The Layman Analogy



Using a master neural network to expand simple binary switches into a rich, high-fidelity, blendable universe.

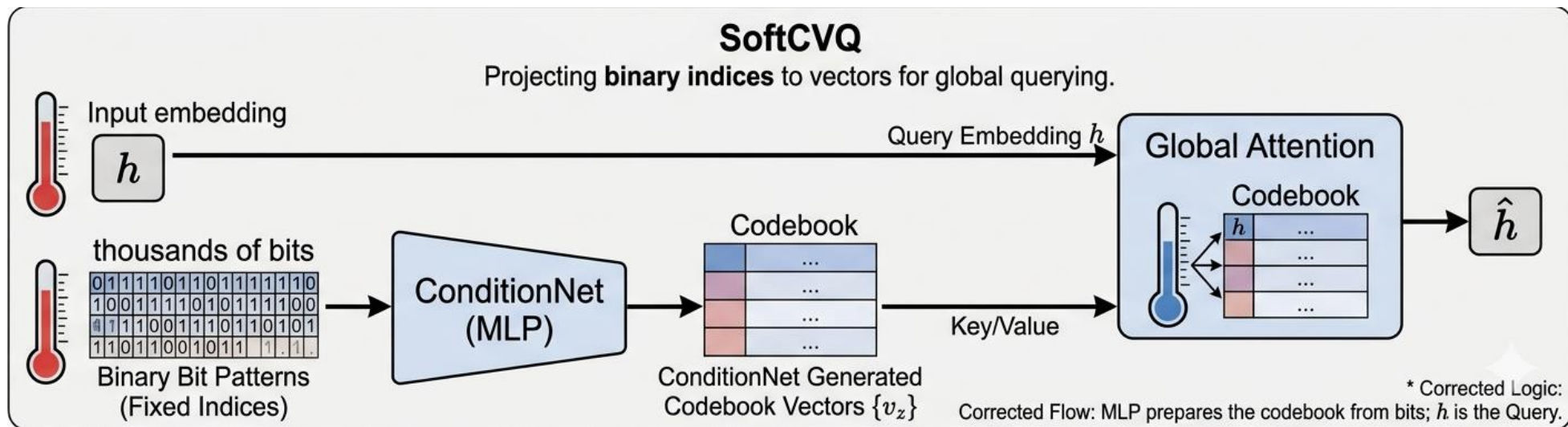
Technical Definition

The Ultimate Synthesis

Uses an MLP ConditionNet to mathematically project low-dimensional binary vectors into a massive high-dimensional continuous space.

Enables flawless Soft Global Querying for reconstruction while natively retaining binary VQ-IDs for generative stability.

VQ Evolution V: Soft Conditional Vector Quantizer (SoftCVQ)



Mathematical Derivation

$$v_j = \text{ConditionNet}(\text{Bit}(z, \log_2 m))$$

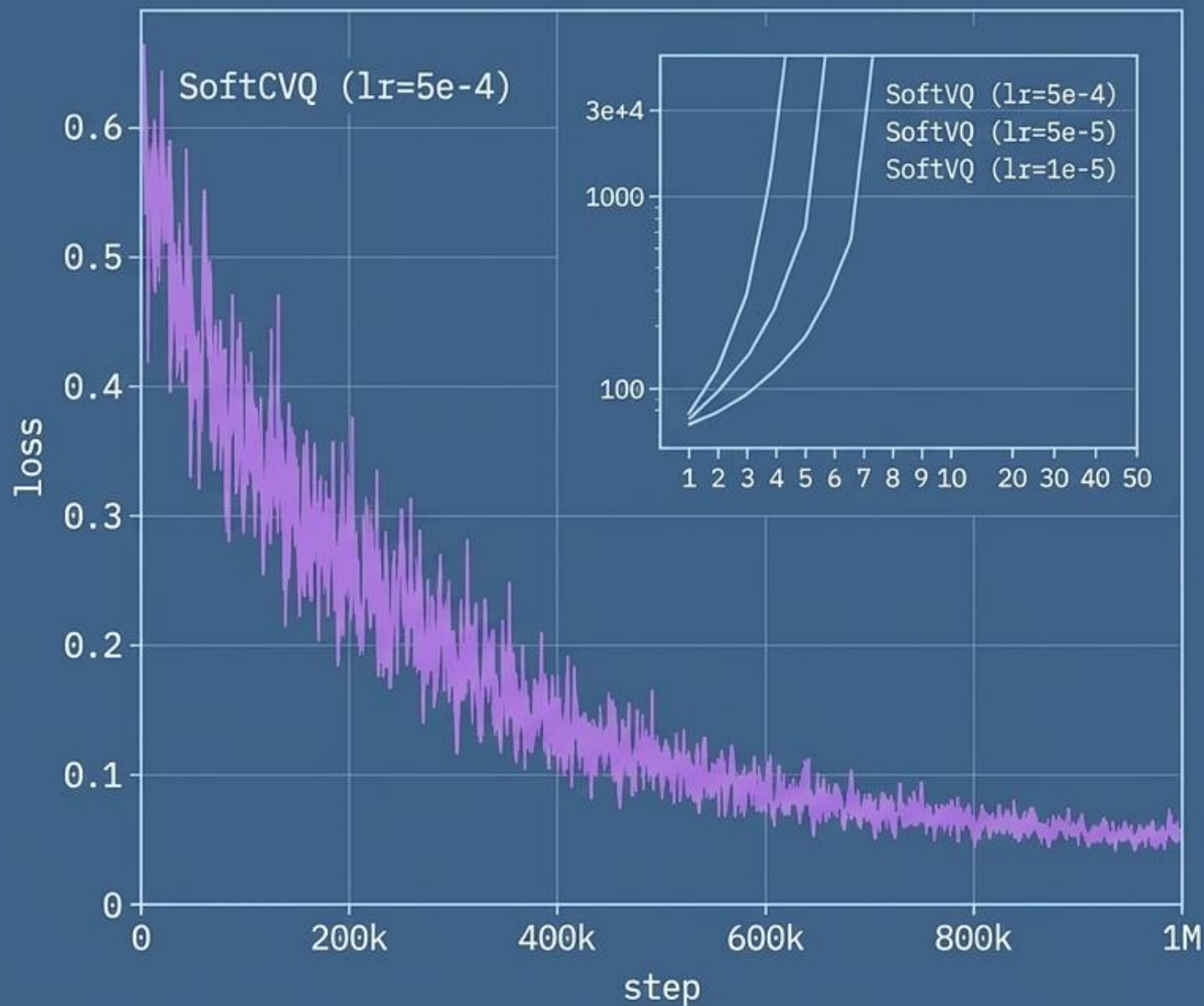
$$\hat{h}_i = \text{SoftVQ}(h_i, \{v_z\})$$

The Discretization Matrix: Solving the VQ Bottlenecks

	w/o Gradient Mismatch	w/o Semantic Irrelevance	w/o Multiclass Classification	w/o Info Bottleneck	WITH Soft Global Querying
Vanilla VQ				✓	
Lookup-Free VQ		✓	✓		
SoftVQ	✓			✓	✓
SoftGVQ	✓	✓	✓	✓	
SoftCVQ	✓	✓	✓	✓	✓

SoftCVQ uniquely decouples the codebook dimension from the vocabulary size, enabling high-fidelity 3D reconstruction without sacrificing sequence generation stability.

Why Binary VQ-IDs Matter for Generation



The Problem with SoftVQ

Using a massive multi-class space (e.g., $2^{16} = 65,536$ distinct IDs) makes generative optimization nearly impossible. Models spike and crash regardless of learning rate.

The SoftCVQ Solution

Decomposes the massive class space into 16 binary subspaces.

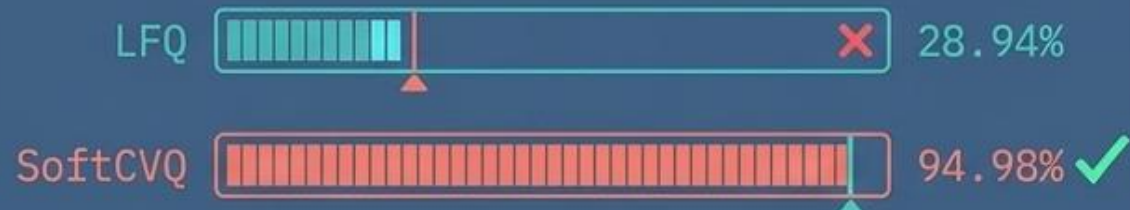
Key Insight

The generation problem shifts from one massive multi-class prediction into 16 robust binary classifications, resulting in smooth, rapid convergence.

Task 1: Information Preservation (Reconstruction)

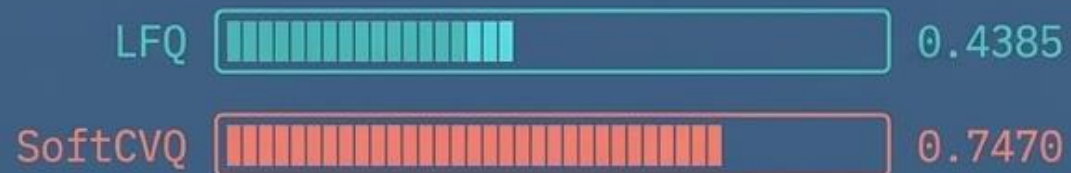
Dataset: CATH4.3 Test Set

Metric 1 (Success Rate)



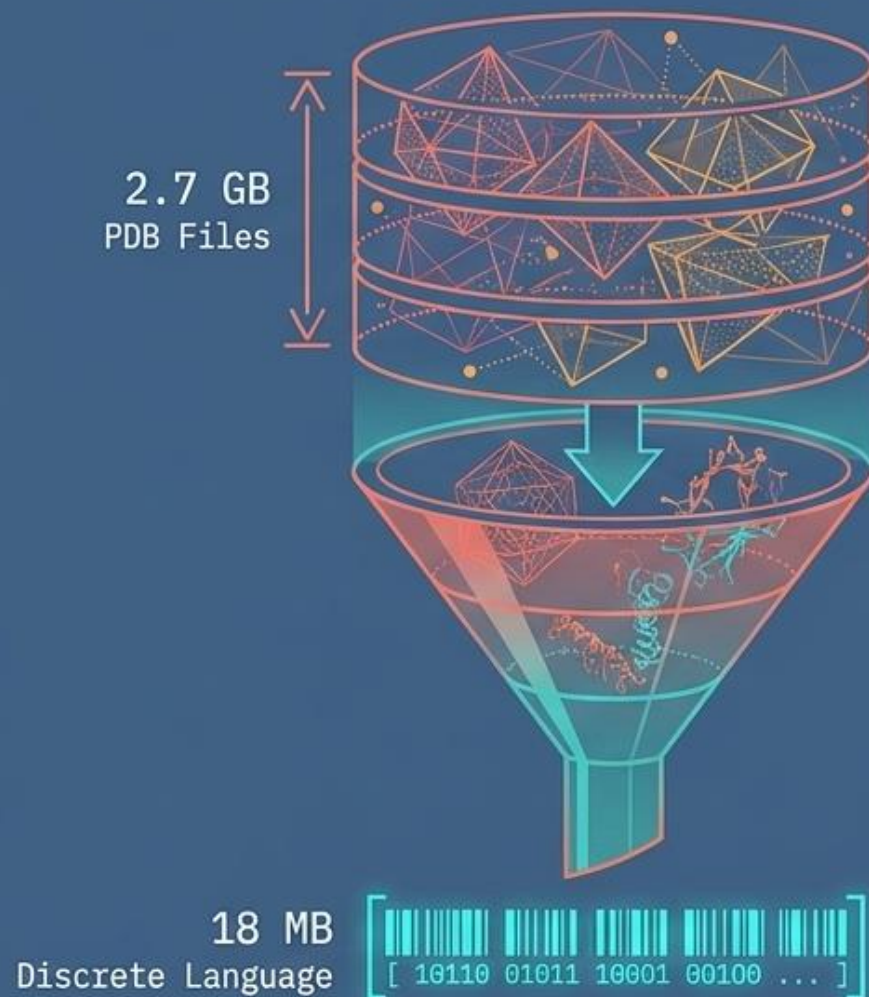
Sequence Reconstruction Accuracy

Metric 2 (TMScore)



Global Structure Similarity (0.0 - 1.0)

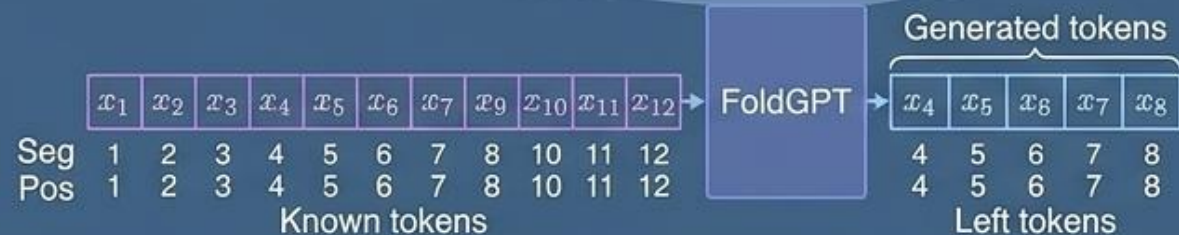
Compression Triumph



SoftCVQ achieves near-perfect sequence recovery, massively compressing continuous shape geometries into 1D without loss.

Stage 2: FoldGPT Architecture

	Known												Key		Unknown				
	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_{10}	x_{11}	x_{12}	x_1	x_2	x_3	x_4	x_6	x_7	x_8	x_9
x_1												×	×	×	×	×	×	×	×
x_2												×	×	×	×	×	×	×	×
x_3													×	×	×	×	×	×	×
x_4														×	×	×	×	×	×
x_5															×	×	×	×	×
x_6																×	×	×	×
x_7																	×	×	×
x_8																		×	×
x_{10}																			×
x_{11}																			×
x_{12}																			×



Concept:

The first GPT-style sequence-structure co-generation model.

Mechanism:

Pure autoregressive transformer. Because the modalities are unified into discrete tokens, there is no need for complex diffusion processes or SE(3) equivariance.

$$\max_{\theta} \mathbb{E} \sum_{k \in I_m} \log p_{\theta}(z_k | z_{\text{known}}, z_{<k})$$

Execution: Utilizes full attention over all known context (segment + position encoding), predicting masked regions sequentially like a standard Large Language Model predicting the next word.

Experimental Setup & Datasets

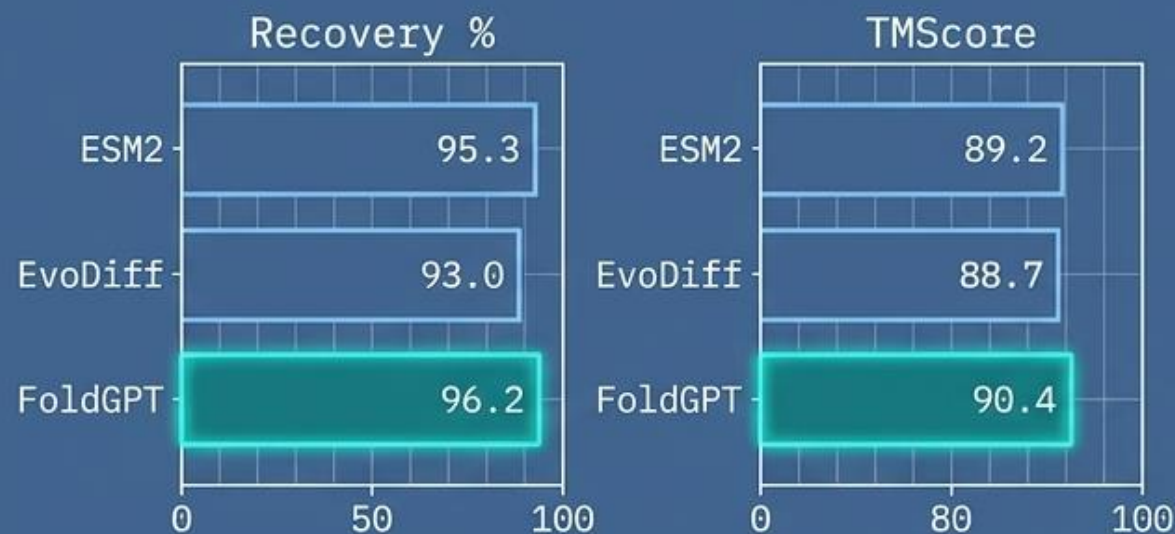
- **VQ Pretraining:** CAF2DB (1.3M proteins from AlphaFold Uniprot v3).
- **Downstream Evaluation: CATH4.3:** 30k training samples for backbone inpainting.
 - **SAbDab:** Structural Antibody Database for CDR design.
- **Config:** ESM2 35M-parameter Transformer, codebook size 2^{16} .

Method	Global Metrics		Success Rates		Local Metrics					
	Rec	TMScore	Rec ≥ 0.95	TMScore ≥ 0.5	L_r	L_α	L_β	max L_r	max L_α	max L_β
Vanilla	<u>0.9757</u>	0.4224	<u>0.9791</u>	0.1693	0.0966	0.0602	0.0783	2.1790	0.5605	0.3723
LFQ	0.9482	0.4385	0.7534	0.2894	0.1066	0.0701	0.1079	2.2068	0.5856	0.5257
SoftVQ (our)	0.9713	0.7616	0.9781	0.9530	<u>0.0683</u>	0.0438	0.0232	<u>1.5647</u>	0.4989	0.1439
SoftGVQ (our)	0.9743	0.5703	0.9875	0.5120	0.0936	0.0616	0.0619	2.0090	0.5655	0.3054
SoftCVQ (our)	0.9880	<u>0.7470</u>	0.9603	<u>0.9498</u>	0.0523	<u>0.0447</u>	<u>0.0284</u>	0.8274	<u>0.5334</u>	<u>0.1646</u>

Table 2: Performance of protein VQ models on CATH4.3 test set. We highlight the **best** results and suboptimal.

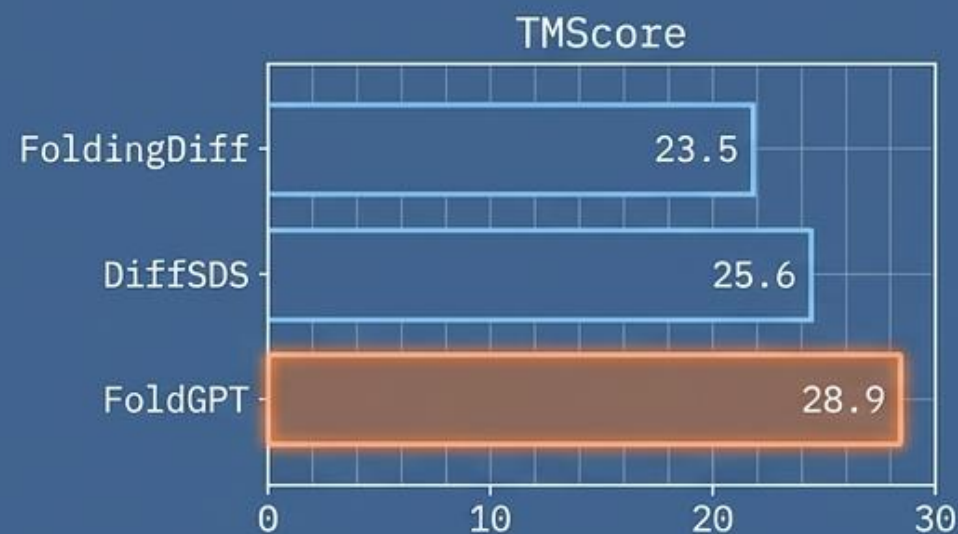
Task 2: Backbone Inpainting Performance

Sequence Design



FoldGPT beats established sequence models despite using significantly less training data.

Structure Design (Angle-based)

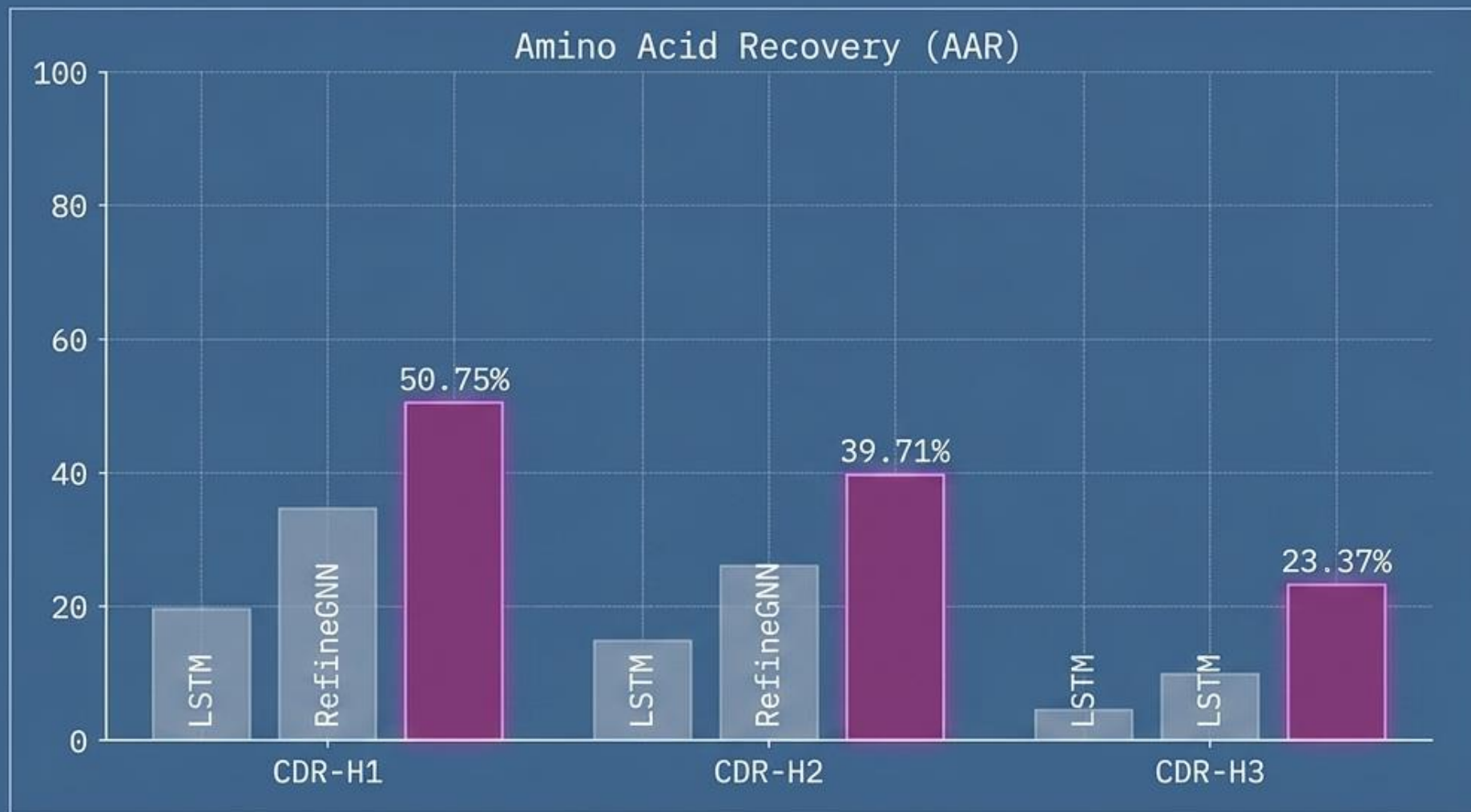


Substantially outperforms other angle-based models.

Crucial Takeaway: Discretizing angles into a unified language performs structurally better than generating continuous angle sequences.

Task 3: Application to Antibody Design

Context: Fine-tuned on the SAbDab database specifically for targeted CDR design.



FoldGPT systematically outperforms other antibody-only baselines across all CDR loops.

Significance: Proves the immediate practical utility of the 'Protein Language' in complex, real-world therapeutic generative scenarios.

	Sequence Design			Structure Design (angle-based)			Structure Design (coordinate-based)	
	$\text{Rec}(\mathcal{S}, \hat{\mathcal{S}})$	$\text{TM}(\mathcal{X}, \hat{\mathcal{X}}')$		$\text{TM}(\mathcal{X}, \hat{\mathcal{X}})$	$\text{Rec}(\mathcal{S}, \hat{\mathcal{S}}')$		$\text{TM}(\mathcal{X}, \hat{\mathcal{X}})$	$\text{Rec}(\mathcal{S}, \hat{\mathcal{S}}')$
ESM2	95.3	89.2	FoldingDiff	0.67	23.5	ProtDiff	0.95	28.4
EvoDiff	93.0	88.7	DiffSDS	0.74	25.6	RFDiffusion	0.99	33.9
FoldGPT	96.2	90.4	FoldGPT	0.80	28.9	Chroma	0.98	34.5

Table 3: Results of backbone inpainting.

Visualizing the Blueprint: Sequence-Structure Recovery

Direct structural overlays demonstrate the unprecedented geometric fidelity maintained by the 1D discrete FoldToken framework.



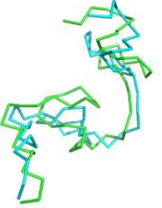
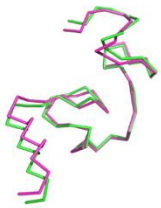
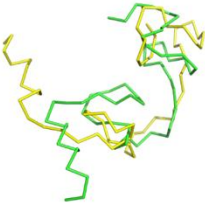
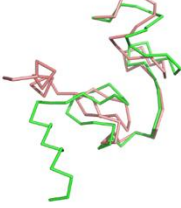
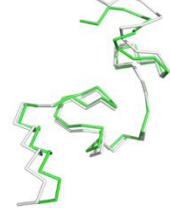
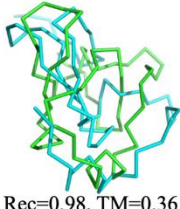
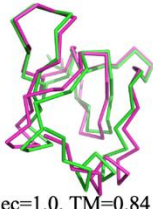
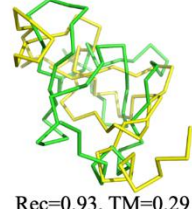
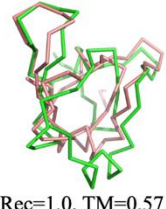
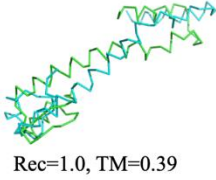
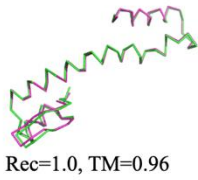
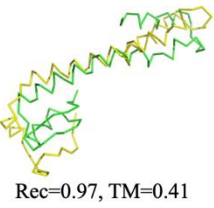
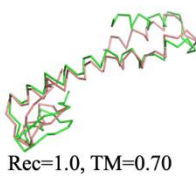
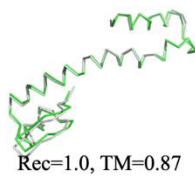
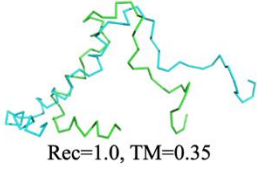
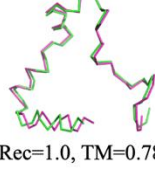
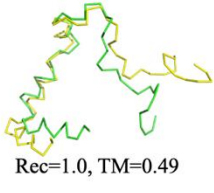
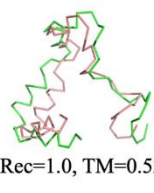
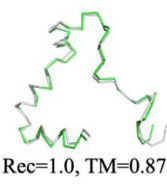
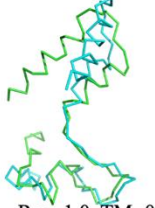
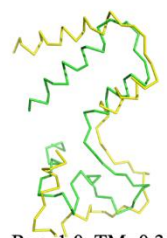
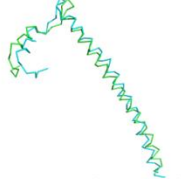
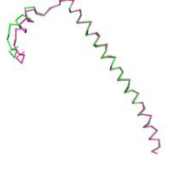
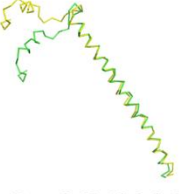
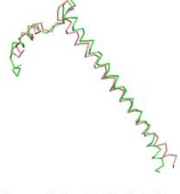
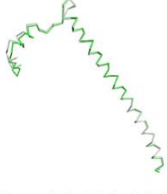
PDB 1a1tA00: Rec=0.98, TM=0.77



PDB 2kz6A01: Rec=1.0, TM=0.87



PDB 5ca5A03: Rec=1.0, TM=0.93

PDB Name	Vanilla VQ	SoftVQ	LFQ	SoftGVQ	SoftCVQ
1altA00	 Rec=1.0, TM=0.35	 Rec=0.98, TM=0.78	 Rec=0.98, TM=0.29	 Rec=1.0, TM=0.51	 Rec=0.98, TM=0.77
1isuA00	 Rec=0.98, TM=0.36	 Rec=1.0, TM=0.84	 Rec=0.93, TM=0.29	 Rec=1.0, TM=0.57	 Rec=1.0, TM=0.63
2kz6A01	 Rec=1.0, TM=0.39	 Rec=1.0, TM=0.96	 Rec=0.97, TM=0.41	 Rec=1.0, TM=0.70	 Rec=1.0, TM=0.87
3i9v101	 Rec=1.0, TM=0.35	 Rec=1.0, TM=0.78	 Rec=1.0, TM=0.49	 Rec=1.0, TM=0.52	 Rec=1.0, TM=0.87
4iaoC01	 Rec=1.0, TM=0.39	 Rec=1.0, TM=0.76	 Rec=1.0, TM=0.34	 Rec=1.0, TM=0.50	 Rec=1.0, TM=0.79
5ca5A03	 Rec=1.0, TM=0.64	 Rec=1.0, TM=0.90	 Rec=0.99, TM=0.65	 Rec=1.0, TM=0.70	 Rec=1.0, TM=0.93

Method-ID	Config		Sphere	Global Metrics		Success Rates	
	#MLP	#CodeDim		Rec	TMScore	Rec ≥ 0.95	TMScore ≥ 0.5
SCQ-1	2	16		0.9801	0.5853	0.8997	0.7638
SCQ-2	2	16	✓	0.9879	0.6451	0.9572	0.8892
SCQ-3	2	32		0.9965	0.4285	0.9937	0.2184
SCQ-4	2	32	✓	0.9896	0.6959	0.9561	0.9216
SCQ-5	3	32		0.9875	0.5867	0.9457	0.7367
SCQ-6	3	32	✓	0.9862	0.6697	0.9467	0.9028
SCQ-7	6	32	✓	0.9880	0.7470	0.9603	0.9498
SCQ-8	9	32	✓	0.9875	0.7015	0.9551	0.9101

Table 5: Ablation of SoftCVQ on CATH4.3 test set.

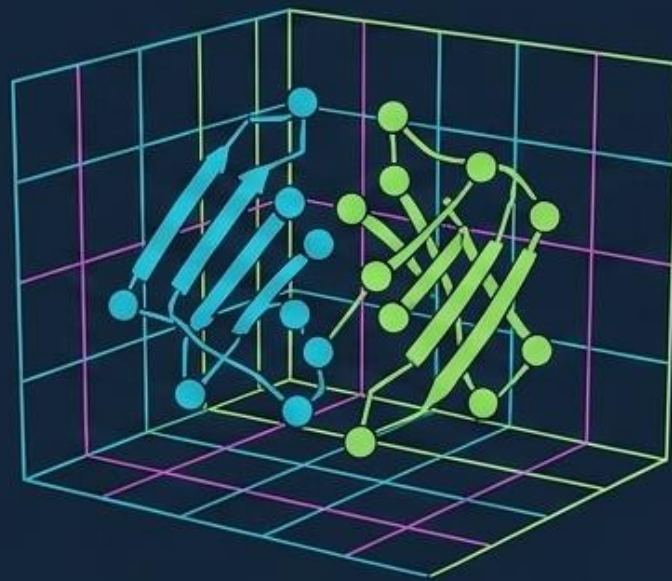
Limitations & The Path Forward

Angle-Based Error Accumulation



Current limitation: **Angle-based structure representation** risks sequential error accumulation and struggles to accurately model complex 3D pairwise interactions, such as **antibody-antigen complexes**.

Coordinate-Based Modelling



The Path Forward: The next evolution will adapt **FoldToken** directly for **coordinate-based modelling**, allowing precise, flawless 3D generation of **multi-chain complexes**.

The New Paradigm in Generative Biology

1. Unified Modality: Sequence and structure are no longer divided; they are a single, universally spoken discrete language.

The "Rosetta Stone"
fully assembled Cont

2. SoftCVQ: The vector quantisation paradox is solved, enabling flawless compression combined with highly robust generation.

>SEQ: MKTIIALSYIFCL...

...

[FOLD_TOKEN:CYAN:001]

...

[FOLD_TOKEN:CYAN:001]

...

[FOLD_TOKEN:CYAN:001]

```
[0101001010 010000001110
0101001011 001010110100
0101001101 011010000100
0101010010 ...]
```



3. GPT for Biology: FoldGPT proves that language models can natively invent physical, 3D biological molecules from scratch.



The modality chasm is bridged. The era of pure language-driven protein co-design has arrived.